

Big-Data-Seminar

Tipps und Hinweise

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Goal of the seminar

- Learn how to
 - **Handle** a scientific paper
 - **Present** a scientific paper
 - **Summarize** scientific work

Goal of this talk

- Tips and hints
 - Organization
 - Delivery

1. Motivation
2. Organization of a presentation
3. Delivery
4. Further reading

Communicating knowledge

- Problem
 - Communication failure
- Reason
 - **Written** $\neq$ **oral** communication
 - Know your audience
- Focus
 - Reading vs. presenting
 - Presenting vs. writing

Presenting a paper

- Target: The audience (**not** you!)
- What you want to present
 - Motivation
 - Problem background
 - Problem statement
 - Proposed solution
 - Method
 - Biological background
 - Computer science background
 - Algorithm used
 - Results
 - Summary
- Clear and terse messages

Example: Prosit - proteome-wide prediction of peptide mass spectra by deep learning
peptide mass spectra ← Biological background

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peptide mass spectra ← **Biological background**

prediction by deep learning ← **Computer science background**

Example:

*In mass-spectrometry-based proteomics, the **identification and quantification of peptides** ← **Task** and proteins heavily rely on sequence database searching or spectral library matching. The lack of accurate predictive models for fragment ion intensities impairs the realization of the full potential of these approaches. [...] We show the general applicability of Prosit by predicting spectra for proteases other than trypsin, generating spectral libraries for data-independent acquisition and improving the analysis of metaproteomes. [...]*

Example:

*In mass-spectrometry-based proteomics, the identification and quantification of peptides and proteins heavily rely on sequence database searching or spectral library matching. **The lack of accurate predictive models** ← **Problem** for fragment ion intensities impairs the realization of the full potential of these approaches. [...] We show the general applicability of Prosit by predicting spectra for proteases other than trypsin, generating spectral libraries for data-independent acquisition and improving the analysis of metaproteomes. [...]*

Example:

*In mass-spectrometry-based proteomics, the identification and quantification of peptides and proteins heavily rely on sequence database searching or spectral library matching. The lack of accurate predictive models for fragment ion intensities impairs the realization of the full potential of these approaches. [...] We show the general applicability of Prosit **by predicting spectra for proteases** ← **Solution** other than trypsin, generating spectral libraries for data-independent acquisition and improving the analysis of metaproteomes. [...]*

Organization of a presentation

Effective presentation

- Organize your thoughts
 - Brainstorm: What do I want to tell my audience?
- Structure the presentation
 - Organize the information for the audience
 - Retain the logical interconnections
 - Your audience might not be familiar with the paper or topic!
- Practice presenting
 - A good talk is a result of **experience** and **practice**
 - Practice your talk at least **five to ten** times!

Structure: The rule of three

1. Tell them what you are about to tell
 - Introduction
2. Tell them
 - Body
3. Tell them what you have told
 - Conclusion

Structure: Introduction

1. Tell them what you are about to tell
 - Introduction
 - Title and authors; introduce yourself (1 slide)
 - Overview of the presentation (1 slide)
2. Tell them
 - Body
3. Tell them what you have told
 - Conclusion

Organization of a presentation

Structure: Body

1. Tell them what you are about to tell
 - Introduction
2. Tell them
 - Body
 - Motivation (2–3 slides)
 - Methods (4–6 slides)
 - Results (2–3 slides)
3. Tell them what you have told
 - Conclusion

Organization of a presentation

Structure: Conclusion

1. Tell them what you are about to tell
 - Introduction
2. Tell them
 - Body
3. Tell them what you have told
 - Conclusion
 - Summary (1–2 slides)
 - References (1 slide)
 - Backup slides (any number)

Slides

- Timing
 - Time per slide: 1-2 minutes $\Rightarrow$ ca. 20 slides
- Text
 - Avoid walls of text and full sentences
 - **Highlight text** (e.g., messages)
 - Use different colors and fonts carefully
 - Do not show information that you will not explain
- Visuals
 - A figure is worth a thousand words!
 - Explain your figures: The reader will not guess
 - Use tables **very** carefully

Delivery

- Face your audience
- Speak loudly and clearly
- Speak freely
 - Do not read from your slides
 - **Do not bring notes**
- Do not rush through the slides
 - Filter (backup slides)
 - Practice to finish on time
- Focus on clear messages
- Use the rule of three

Further reading

- How to read a scientific article

<http://faculty.fiu.edu/~collinsl/Article%20reading%20tips.htm>

- How to present a paper

<https://cc.gatech.edu/faculty/ashwin/wisdom/how-to-present-a-paper.html>

- Oral presentation advice

<http://pages.cs.wisc.edu/~markhill/conference-talk.html>

- Pointers on giving a talk

https://people.eecs.berkeley.edu/~messer/Bad_talk.html

- Hilary Glasman-Deal. Science Research Writing. Imperial College Press, London. 2010.